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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

MVJ22ECP76-PROJECT PHASE-2 – SECOND REVIEW

AUTONOMOUS MULTITASKING RESCUE ROBOT IN UNMANNED ZONES

Presented by,

1MJ22EC042	Deepali Dinesh B
1MJ22EC051	Ganavi V R
1MJ22EC054	Geetha M K
1MJ22EC059	Hamsa K

Guided by,

Dr. Chippy M Manu
Assistant Professor,
Dept. of ECE

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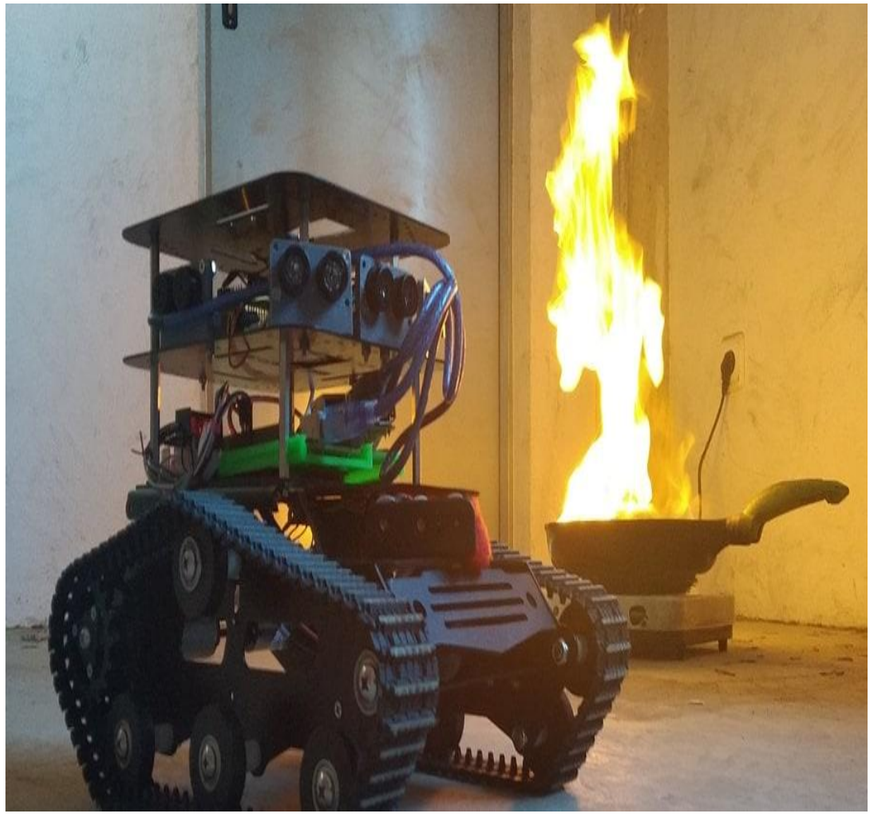
ABSTRACT

- In disaster-stricken environments such as fire zones, gas leaks, and building collapses, human intervention for rescue operations can be highly dangerous and life-threatening.
- To address this challenge, we propose a Multitasking Rescue Robot capable of detecting environmental hazards, locating victims, assessing their health condition, and transmitting real-time data to a remote operator.
- The model analyzes the data and predicts the person's health condition, classifying it as normal, mild risk, or critical, thus allowing rescuers to prioritize actions accordingly.

INTRODUCTION

- Disasters create dangerous environments for humans.
- Human rescuers face high risk in such conditions.
- There is a need for autonomous rescue systems.
- Robots can reduce human risk in hazardous areas.
- They help in hazard detection, victim location, and health assessment



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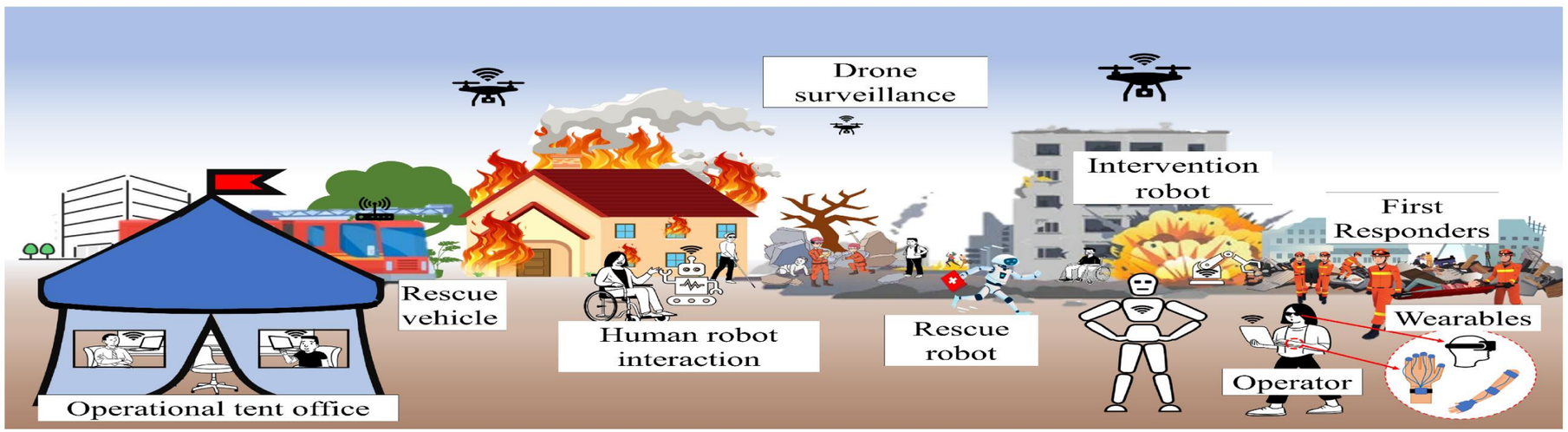
PROBLEM STATEMENT

Rescue operations need a safe robot capable of entering unstable disaster zones to detect threats and locate people in need of help.



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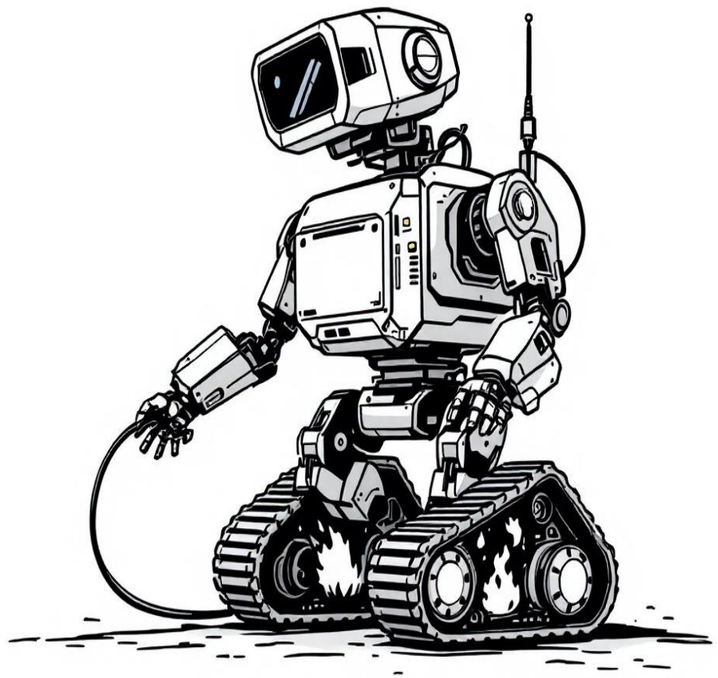
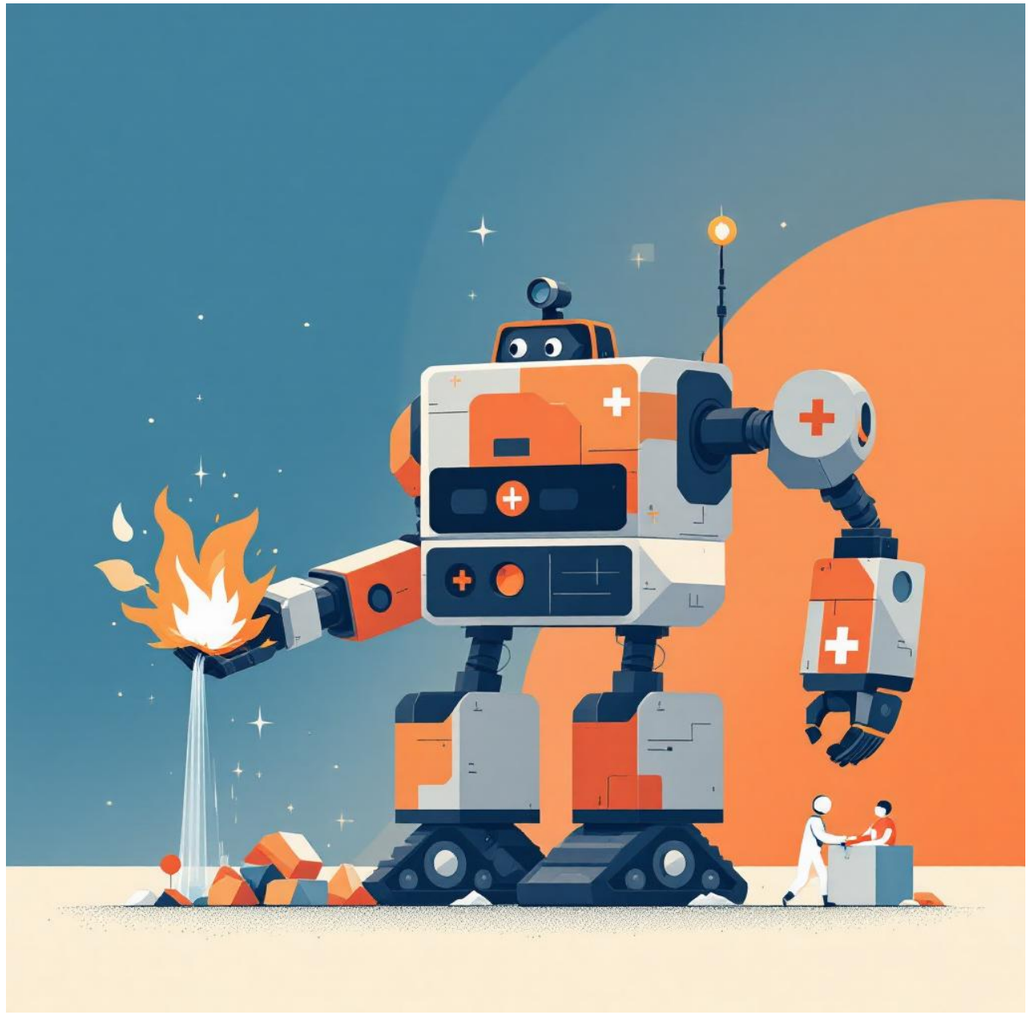
OBJECTIVES

- To reduce human risk in dangerous rescue environments
- To overcome delays in victim detection and medical assessment
- To leverage modern low-cost technology for smarter rescue operations



MOTIVATION

- To reduce the risk for rescuers
- To identify who needs help first using machine learning..



LITERATURE SURVEY

01.“Object Identification, Localization and Monitoring Framework for Smart Mobility Application”

Published by:IEEE Transactions on Intelligent Transportation Systems

Published on: 12 March 2024

DOI: 10.1109/TITS.2022.3148752

Authors: Zhihao Chen, Yuxin Zhang, and Wei Liu

High computational load-difficult to run on low-power devices.

Limited testing on real-world traffic or public mobility systems.

Accuracy reduces under low-light, fog, or rain conditions.

LITERATURE SURVEY

02. "Object Detection with Deep Learning: A Review

Published by: IEEE Transactions on Neural Networks and Learning Systems

Published on: 21 February 2024

DOI: 10.1109/TNNLS.2019.2892409

Authors: Zhong-Qiu Zhao, Peng Zheng, Shou-tao Xu, and Xindong Wu

Lack of coverage of newer or domain-specific datasets.

Many methods require large training data, which the survey acknowledges as a drawback.

High dependency on GPU/TPU hardware for deep learning models.

Most algorithms struggle with small or overlapping objects.

LITERATURE SURVEY

03. "Deep Learning in Object Detection: A Comprehensive Survey"

Published by: IEEE Access

Published on: 24 June 2025

DOI: 10.1109/ACCESS.2019.2926510

Authors: Licheng Jiao, Fan Zhang, Fang Liu, Shuyuan Yang, and Lingling Li

Models require huge computational resources for training.

Difficulty in generalizing across different domains (e.g., drone images vs. CCTV).

Lacks focus on real-time edge deployment.

LITERATURE SURVEY

04. "A Hybrid CNN–SVM Model for Object Detection in Autonomous Driving"

Published by: Elsevier – Expert Systems with Applications

Published on: 10 October 2025

DOI: 10.1016/j.eswa.2021.115982

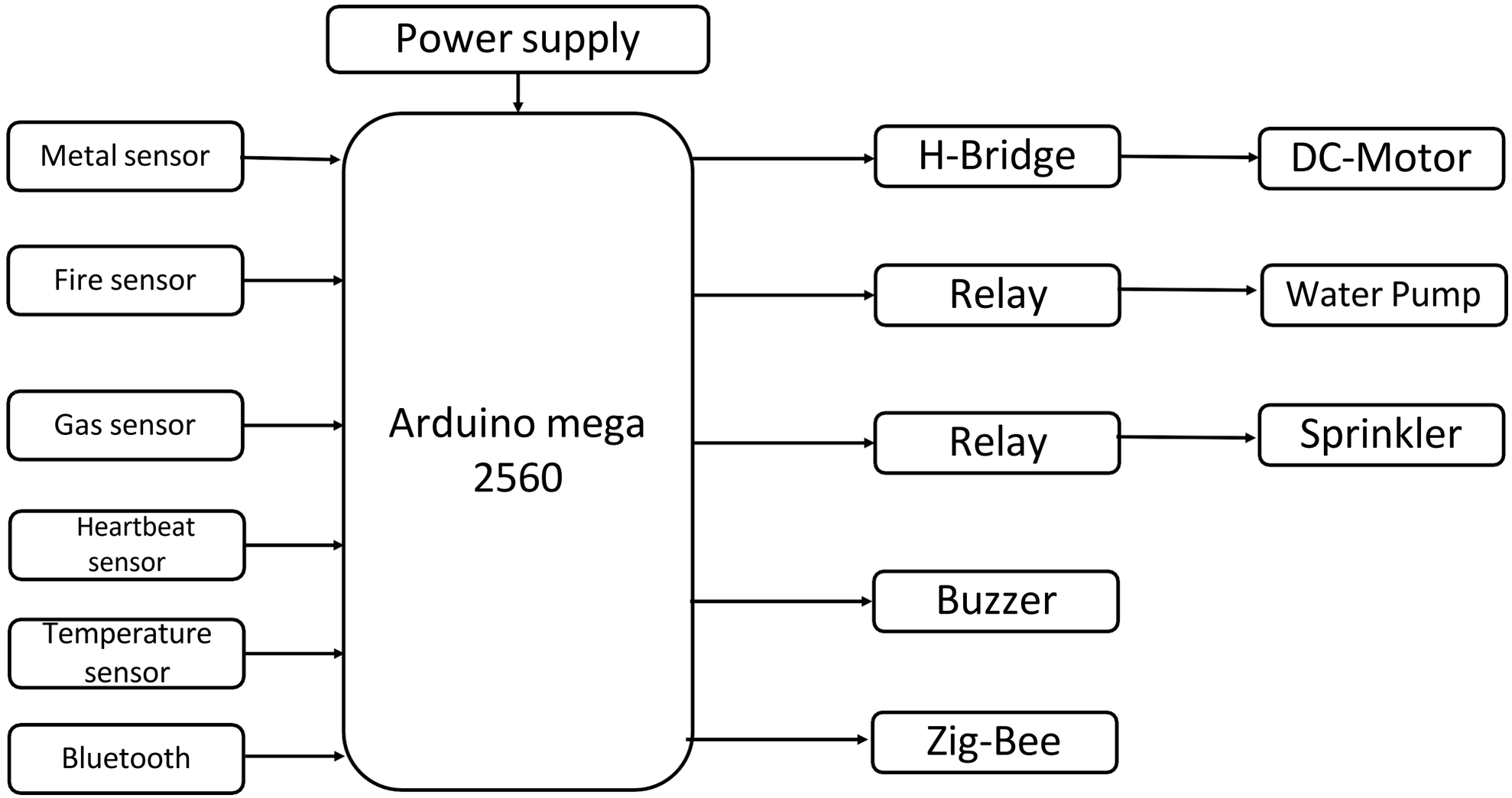
Authors: Yakup Demir, Mehmet Can, and Ali Osman

Performance decreases when vehicles move at high speeds or in dynamic environments.

SVM classifier struggles with very large data sets.

Limited robustness to night-time images or bad weather.

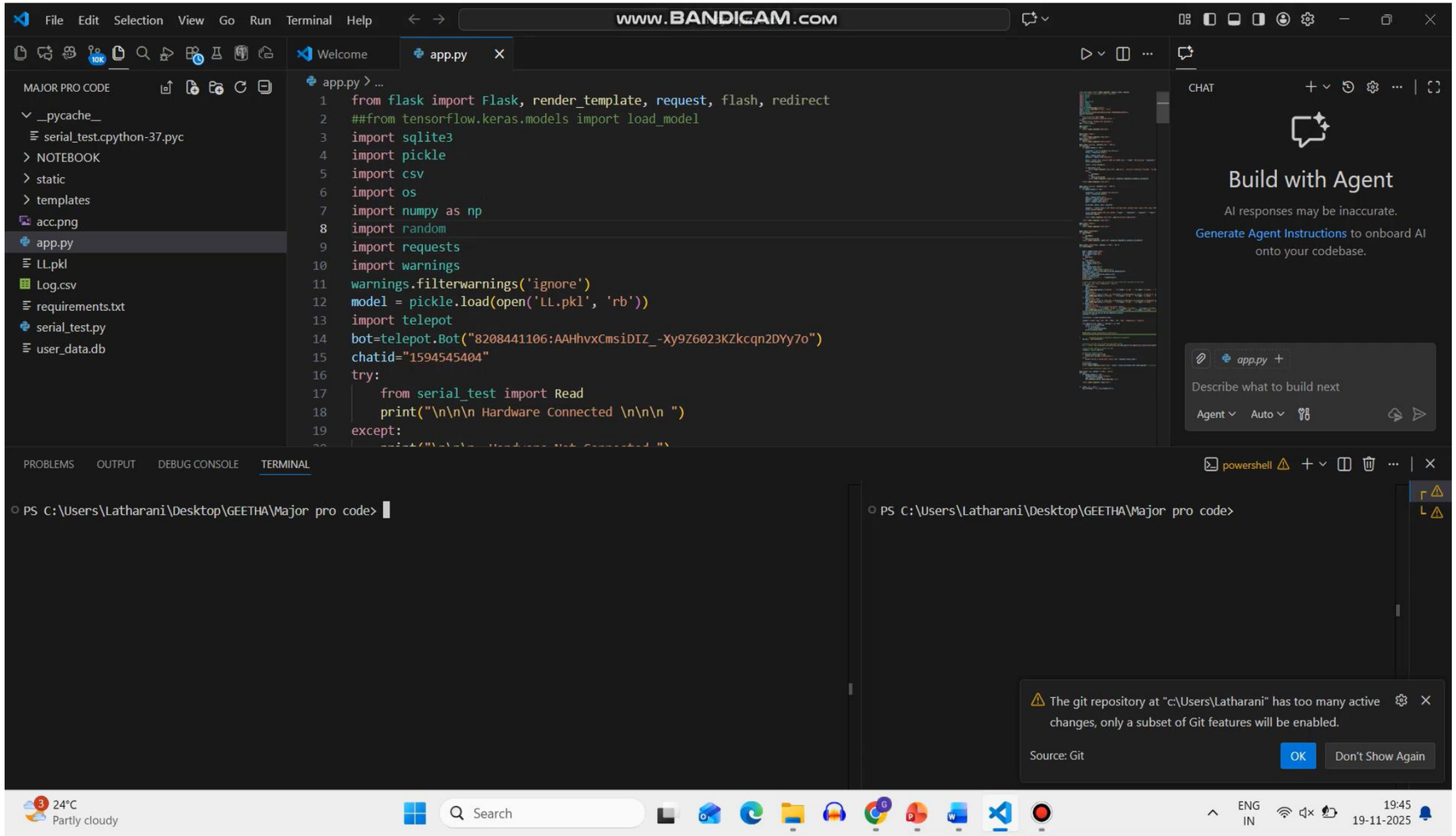
BLOCK DIAGRAM



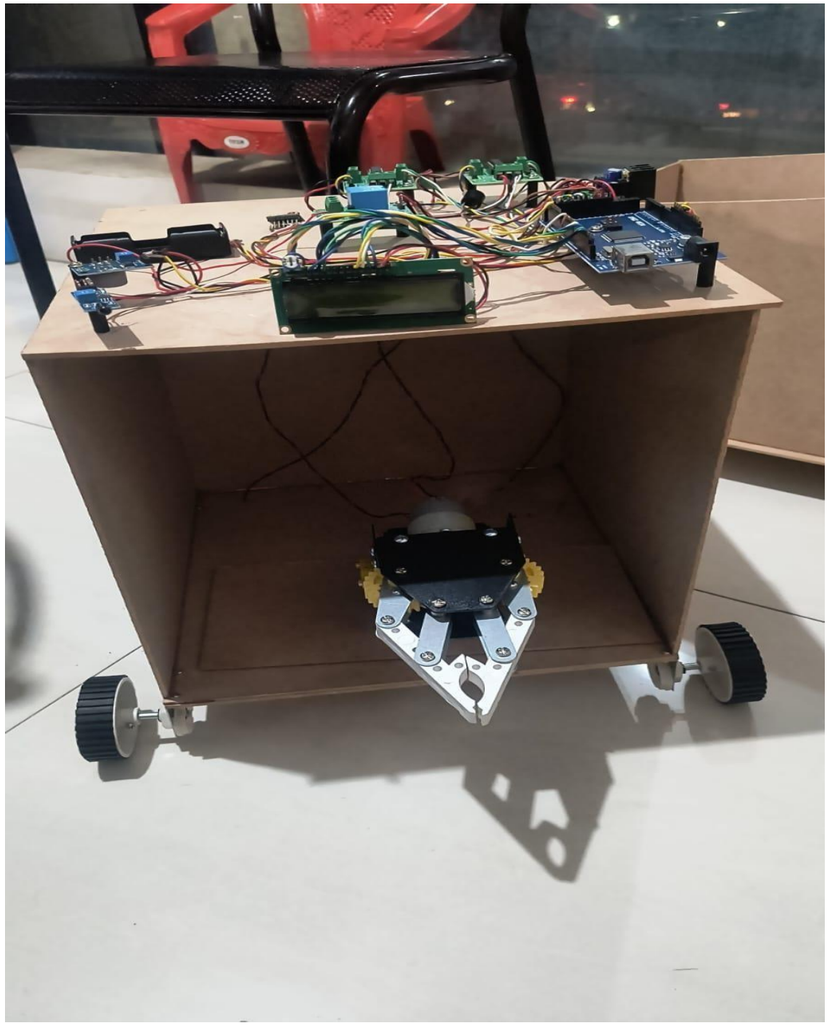
RESULT AND ANALYSIS

The robot moved autonomously and avoided obstacles while detecting fire, gas leaks, and metal objects. It monitored the victim's heartbeat and temperature and sent all data in real time through Bluetooth and Zigbee. A robotic arm handled basic pick-and-place rescue tasks. The machine learning model predicted the victim's health condition. Overall, the system combined multiple rescue functions into one efficient platform.

RESULT AND ANALYSIS



PROTOTYPE



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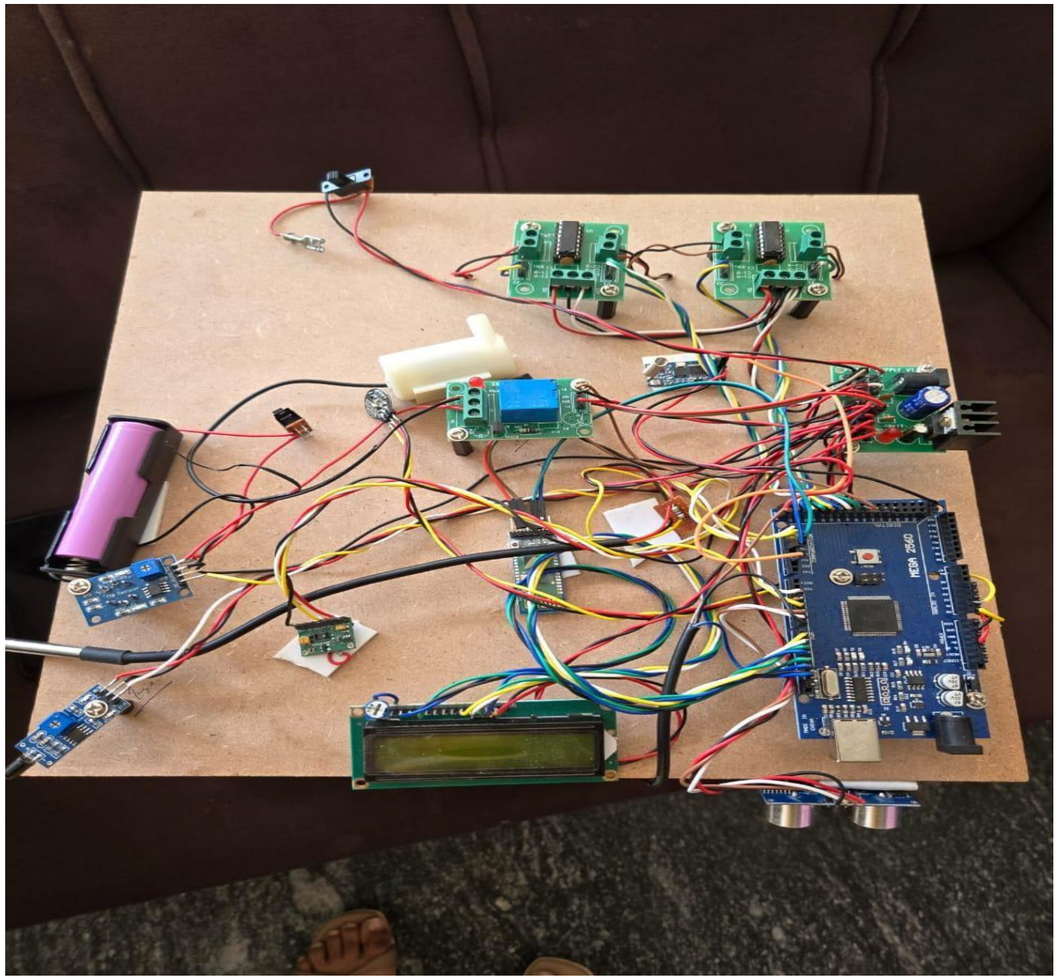

Fig No:6

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THANK YOU